Claim: Seq 9 $\longrightarrow \lim_{n\to\infty}\left\{\frac{1}{2n}\left[1-(-1)^n\right]-\frac{1}{2(n-1)}\left[1+(-1)^n\right]\right\}=0$

Proof: Take any arbitrary positive number ε.

Let, $\forall K\in\mathbb{R}$, $\left|\frac{1}{2K}\left[1-(-1)^K\right]-\frac{1}{2(K-1)}\left[1+(-1)^K\right]-0\right|>\varepsilon$

$\left|\frac{1}{2K}\left[1-(-1)^K\right]-\frac{1}{2(K-1)}\left[1+(-1)^K\right]\right|>\varepsilon$. Let K be odd.

$\therefore \left|\frac{1}{2K}\left[1-(-1)\right]-\frac{1}{2(K-1)}\left[1+(-1)\right]\right|>\varepsilon \Rightarrow \left|\frac{1}{K}\right|>\varepsilon \Rightarrow \frac{1}{\varepsilon}>K$

This is a contradiction since the set of real numbers has no upper bound.

Let K be even. $\therefore \left|\frac{1}{2K}\left[1-(1)\right]-\frac{1}{2(K-1)}\left[1+(1)\right]\right|>\varepsilon \Rightarrow \left|-\frac{1}{K-1}\right|>\varepsilon$

$\Rightarrow \frac{1}{K-1}>\varepsilon \Rightarrow \boxed{\frac{1}{\varepsilon}+1>K}$. This is a contradiction since the set of real numbers has no upper bound.

$\therefore \exists K\in\mathbb{R}$ s.t. $\left|\frac{1}{2K}\left[1-(-1)^K\right]-\frac{1}{2(K-1)}\left[1+(-1)^K\right]-0\right|<\varepsilon$

Like proofs similar to before we know $\forall x\in\mathbb{R}$, $x\geq K$,

$$\boxed{\left|\frac{1}{2x}\left[1-(-1)^x\right]-\frac{1}{2(x-1)}\left[1+(-1)^x\right]-0\right|<\varepsilon} \quad \square$$

Reader: But couldn't the proof of the equality $\lim_{n\to\infty}\frac{n}{n+1}=1$ be simplified?

Author: Have a try.

Reader: Well, first I rewrite the expression in the following way: $\lim_{n\to\infty}\frac{n}{n+1}=\lim_{n\to\infty}\frac{1}{1+\frac{1}{n}}$. Then I take into consideration that with an increase in n, fraction $\frac{1}{n}$ will tend to zero, and consequently, can be neglected against unity.

Hence, we may reject $\frac{1}{n}$ and have: $\lim_{n\to\infty}\frac{1}{1}=1$.